

Pranitha Seemalamudi

AI/ML Engineer

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Professional Summary:

AI/ML Engineer with 3+ years of experience designing, deploying, and scaling production-grade machine learning solutions in cloud environments. AWS Certified Machine Learning Engineer Associate, with proven expertise in building computer vision and NLP applications for healthcare and enterprise domains. Combines strong data engineering capabilities with MLOps best practices to deliver secure, high-performance AI systems that drive operational efficiency and clinical decision-making. Skilled at translating complex model outputs into actionable business strategies and collaborating with cross-functional stakeholders to solve real-world problems.

Technical Skills:

- **Programming Languages:** Python, R, SQL, Bash/Shell Scripting
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, CatBoost, Ensemble Methods
- **Deep Learning:** TensorFlow, Keras, PyTorch, CNNs, RNNs, LSTMs, GANs
- **NLP & LLMs:** Hugging Face Transformers, BERT, GPT, RoBERTa, T5, spaCy, NLTK, LangChain, RAG
- **Data Science & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Tableau, Looker
- **Big Data & ETL:** Apache Spark (PySpark), Hadoop, Hive, Kafka, Airflow, Databricks
- **Databases:** PostgreSQL, MongoDB, MySQL, DynamoDB, Redis, Elasticsearch
- **Cloud Platforms:** AWS (SageMaker, S3, EC2, Lambda, Glue, Redshift, RDS, CloudFormation), Azure
- **MLOps & Deployment:** Docker, Kubernetes, Jenkins, MLflow, Kubeflow, FastAPI, Flask, Streamlit, Terraform
- **DevOps & Tools:** Git, GitHub, GitLab CI/CD, JIRA, Confluence, Linux, Windows, VS Code
- **Statistical Analysis:** Hypothesis Testing, Regression Analysis, Time Series Analysis, A/B Testing
- **Model Monitoring:** Prometheus, Grafana, Evidently AI, AWS CloudWatch

Work Experience:

Mass General Brigham | AI/ML Engineer | Somerville, MA | Jun 2025 – Present

- Architected a PyTorch-based deep learning pipeline for MRI and CT scan classification, decreasing average diagnostic review time by 15% through automated anomaly detection.
- Fine-tuned a BERT-based NLP model to extract key clinical indicators from unstructured physician notes, achieving 0.89 precision and enabling faster patient data retrieval for downstream applications.
- Deployed an automated CI/CD pipeline using Jenkins and Docker, integrating models into AWS SageMaker to streamline updates and reduce production deployment time by 40%.
- Built a secure, HIPAA-compliant data ingestion framework using Apache Kafka and AWS Glue, processing over 5TB of high-volume imaging and EHR data for real-time analytics.
- Designed and launched RESTful APIs with FastAPI to serve model predictions, ensuring sub-50ms response times for clinical dashboard integrations.
- Executed rigorous hyperparameter tuning and cross-validation on AWS SageMaker, improving model generalization by 20% across multiple healthcare datasets.
- Integrated MLflow for comprehensive experiment tracking, model versioning, and performance monitoring, ensuring full reproducibility and auditability of all ML assets.
- Engineered a PostgreSQL database schema optimized for storing model metadata, patient outcomes, and feature stores, enabling rapid data retrieval for research queries.
- Performed SHAP-based feature importance analysis to enhance model interpretability, building trust with healthcare professionals and regulatory reviewers.
- Collaborated with clinical researchers to define performance metrics (AUC-ROC, Sensitivity, Specificity) and translate model outputs into actionable clinical protocols.

Environment: AWS (SageMaker, S3, EC2, Lambda, Glue, CloudWatch), PyTorch, BERT, FastAPI, Docker, MLflow, PostgreSQL, Apache Kafka, Python, Pandas, NumPy, SHAP.

University of North Texas | AI/ML Research Assistant | Denton, TX | Aug 2024 – May 2025

- Evaluated and optimized CNN architectures including ResNet and Efficient Net using TensorFlow, achieving a 5% accuracy improvement on limited benchmark datasets through transfer learning.
- Experimented with various data augmentation strategies including rotation, scaling, and flipping to improve model robustness and reduce overfitting on small-scale image datasets.
- Implemented and fine-tuned a sequence-to-sequence model using Hugging Face Transformers for a machine translation project, achieving BLEU score improvements of 3 points over baseline.
- Co-developed a methodology for bias detection and mitigation in NLP models, applying fairness metrics such as demographic parity to ensure ethical AI output.
- Built custom data processing pipelines in Python to handle large-scale text corpora and image datasets, reducing data preparation time by 35%.
- Administered a shared GPU compute cluster, managing job queues and allocating resources to maximize research productivity for a team of 8 graduate students.
- Implemented a robust logging and monitoring system using Weights & Biases to track model performance and system health during large-scale experimental runs.
- Developed a custom data augmentation pipeline using OpenCV and NumPy, increasing the effective training dataset size by 3x and improving model generalization.

Environment: TensorFlow, Keras, PyTorch, Hugging Face Transformers, OpenCV, NumPy, Pandas, Python, Linux, CUDA, Matplotlib, Weights & Biases.

Apollo Hospitals | AI/ML Engineer | India | May 2023 – Jul 2024

- Developed XGBoost and LightGBM models to forecast patient readmission rates with 92% accuracy, enabling proactive care interventions and reducing operational costs by 12%.
- Designed and implemented a centralized AWS Redshift data warehouse, integrating data from 10+ disparate clinical and operational systems into a single source of truth for analytics.
- Built PySpark ETL pipelines to clean, transform, and aggregate over 10 million patient records, increasing data processing speed by 40% and reducing job execution time from hours to minutes.
- Deployed a real-time predictive monitoring system using AWS Lambda and API Gateway, triggering automated alerts for high-risk patients within 2 seconds of data ingestion.
- Created interactive Power BI dashboards to visualize bed occupancy, patient flow, and key operational KPIs, enabling administration to optimize resource allocation and reduce wait times.
- Applied survival analysis and time series forecasting (ARIMA, Prophet) to analyze patient length of stay and treatment efficacy, supporting hospital-wide capacity planning.
- Optimized complex SQL queries and database indexes, cutting report generation time by 25% for business intelligence and executive reporting teams.
- Authored comprehensive technical documentation and data dictionaries, improving cross-functional collaboration and reducing onboarding time for new analysts by 30%.

Environment: AWS (Redshift, S3, Lambda, RDS), PySpark, Scikit-learn, XGBoost, LightGBM, Power BI, SQL, PostgreSQL, Python, Pandas, Git, GitHub.

Education:

- **Master of Science in Data Science** | University of North Texas | Denton, TX
- **Bachelor of Engineering in Electronics and Communication Engineering** | Anna University | India

Certifications:

- **AWS Certified Machine Learning Engineer – Associate** | Amazon Web Services